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TrCLIP-VAD : Weak supervised video anomaly detection by improving CLIP training with text rewriting

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ABSTRACT

The existing video anomaly detection (VAD) methods based on the contrastive language-image pre-training (CLIP) framework primarily rely on spatial-temporal features and label prompts, often overlooking high-level semantic information, such as text descriptions of abnormal behavior. Moreover, while data augmentation techniques are commonly used to expand video frames, text inputs typically remain unchanged throughout the training process, limiting the exposure of different texts to the same image. In this paper, we propose a text rewriting based CLIP model called TrCLIP-VAD. It first generates source caption descriptions for each video and then utilizes the In-Context Learning (ICL) capabilities of large language models (LLMs) to rewrite these captions, thereby enhancing semantic expression. Subsequently, we randomly select a caption and fuse it with visual features to detect anomalies. In addition, we introduce the Local-Global Multi-scale Mamba (LGM-Mamba) module. The LGM-Mamba module aims to capture the local-global temporal dependence of frame sequence and promote the deep fusion of visual and caption features to improve the detection performance. The experimental results show that TrCLIP-VAD achieves state-of-the-art performance on two widely used datasets, XD-Violence and UCF-Crime. Specifically, TrCLIP-VAD achieves 86.05% AP scores on XD-Violence dataset and 88.59% AUC scores on UCF-Crime dataset. The code is released at <https://github.com/vpsg-research/TrCLIP-VAD>.

1. Introduction

Recently, weakly supervised video anomaly detection (WSVAD) has attracted much attention because of its wide range of applications. In the field of public safety, WSVAD can detect potential criminal activities such as shootings or fights, ensuring timely responses and strengthening security in communities. Furthermore, it has potential applications in smart farms, where monitoring devices can intelligently detect suspicious activities such as crop theft and sabotage.

WSVAD aims to train a detection model that uses video-level binary label information (normal and abnormal) to predict the confidence score of frame-level anomalies. Most existing methods follow the process shown in Fig. 1(a), which first utilizes pre-trained models such as Inflated 3D ConvNet (Carreira & Zisserman, 2017) and Temporal Segment Networks (Wang et al., 2016) to extract spatial-temporal visual features. These features are then fed into a binary classifier to obtain an anomaly confidence score, which is used to detect abnormal events. However, such methods are challenging to promote video anomaly de-

tection (VAD) tasks by capturing cross-modal clues of visual-language correlation. In recent years, contrastive language-image pre-training (CLIP) models (Radford et al., 2021) have achieved remarkable success in image-level tasks. Some works (Joo et al., 2023; Lv et al., 2023) attempt to use the CLIP model to detect anomalies, but these methods only utilize visual features extracted from the CLIP's image encoder, overlooking the semantic interplay between vision and text (i.e., caption or label prompt). In contrast, TEVAD (Chen et al., 2023), VadCLIP (Wu et al., 2024b), and TPWNG (Yang et al., 2024) effectively integrate visual and textual features, as shown in Fig. 1(b). For example, TEVAD extracts caption features from videos and fuses them with visual features, leveraging the fused features for anomaly detection. VadCLIP and TPWNG advance fine-grained VAD by pairing label prompts of abnormal categories with visual features, while label prompts cannot provide enough high-level semantic information.

We observe that the caption features can effectively supplement the high-level semantic information lacking in some methods, like VadCLIP and TPWNG. When using the caption and visual

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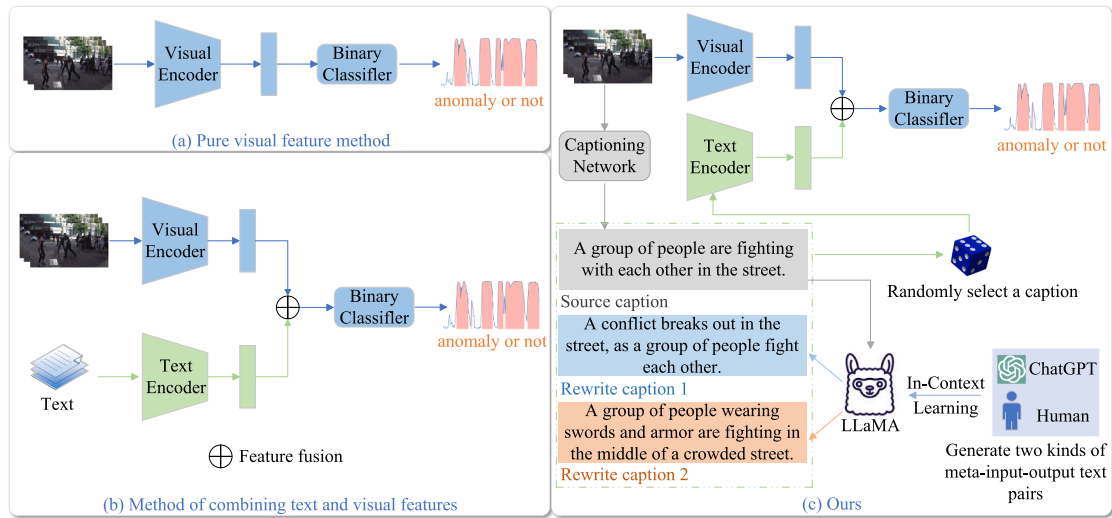


Fig. 1. Comparisons with the existing approaches. Unlike other methods, after obtaining the caption features using the captioning network, we use LLaMA to rewrite the captions into two versions and randomly select one to fuse with visual features for anomaly detection.

features of CLIP directly for training, we face the following two main challenges:

Challenge 1: Unbalanced augmentation between visual and textual modalities. During the CLIP-based training process, data augmentation is widely used to improve model generalization and mitigate overfitting. Visual inputs are typically augmented through transformations such as rotation, cropping, and flipping. However, text input remains unchanged throughout the training process, lacking any form of augmentation. This leads to two problems: Firstly, augmented video frames are always paired with the same captions, leading to image-text data asymmetry. Secondly, the text encoder repeatedly encounters exactly the same caption in each epoch, increasing the risk of overfitting. Unlike image-text rewriting, which focuses on lexical diversity (Fan et al., 2023), video caption rewriting must preserve spatio-temporal causality. For instance, detecting shoplifting requires describing progressive actions (e.g., glancing around → concealing items → hastily exiting), whereas static image captions lack such temporal logic. This fundamental difference necessitates a video-specific rewriting strategy.

Challenge 2: Inconsistency between local detail extraction and global relationship modeling. Due to the continuity of object motion, current events exhibit strong correlations with neighboring events. Meanwhile, anomaly detection requires analyzing their consistency with the global video context over a long time range. Previous methods use classical models in deep learning to capture local or global features separately. CNN-based (Huang et al., 2025; Li et al., 2024a,b; Modi & Parikh, 2024) methods are constrained by limited spatial attention fields, restricting their ability to capture sufficient global features. While Transformer-based (Qu, 2024; Rani et al., 2024; Zhou et al., 2023a) methods achieve global interaction through self-attention mechanisms, traditional self-attention performs fixed computations for all token pairs regardless of their relevance, resulting in computational redundancy. Therefore, how to coordinate the dynamic balance between local feature capture and global spatio-temporal relationship modeling is another key issue of this paper.

To address the two challenges mentioned above, we propose an innovative framework named TrCLIP-VAD. **For the first challenge**, we propose a text rewriting branch as shown in Fig. 1(c), which first generates the source caption for video frame, and then rewrites them sentence by sentence using the In-Context Learning (ICL) ability of LLaMA (Touvron et al., 2023) to enhance the diversity of text expression. ICL is a learning paradigm where models leverage a small number of examples

to understand task requirements, extract key patterns and rules, and use this information to make predictions for new inputs. To leverage LLaMA for text rewriting, we provide two types of rewriting examples (i.e., meta-input-output text pairs): generated by ChatGPT and manually rewritten by humans. After obtaining two kinds of meta-input-output text pairs, the source caption, together with these pairs, is input into the LLaMA model to generate two different caption rewriting texts. In addition, we develop a temporal coherence preservation mechanism to ensure logical event continuity and action causality in caption descriptions. This mechanism focuses on three core pillars: timestamp anchoring, action sequence verification, and causal constraints, as detailed in Section 3.2.2.

For the second challenge, considering that Mamba's selective state space mechanism enables dynamic focus on local features while adaptively suppressing irrelevant frames in long sequences, retaining only critical information to maintain high efficiency. We design the LGM-Mamba module that comprises a local temporal adapter (L), a global temporal adapter (G), a multi-scale temporal network (M), and four Mamba modules. The local temporal adapter primarily captures local temporal dependencies, since current events are highly correlated with adjacent events. In contrast, the global temporal adapter provides a more comprehensive view of feature information with fewer parameters. After obtaining both local and global features, these features are further processed by the multi-scale temporal network (MTN) and the Mamba. Unlike the local and global temporal adapters, Mamba and MTN are capable of capturing contextual information of time series across different scales. As a result, the LGM-Mamba achieves a comprehensive understanding of time series data.

In summary, the primary contributions of our work are as follows:

- We propose a novel TrCLIP-VAD framework, which enhances textual feature diversity through a text rewriting strategy. As far as we know, TrCLIP-VAD is the first work to introduce an image-text dual feature enhancement approach to WSVAD.
- We design an LGM-Mamba module that innovatively integrates the abilities of local perception, global modeling, and multi-scale analysis, enabling efficient and comprehensive temporal feature learning for VAD.
- TrCLIP-VAD achieves state-of-the-art (SOTA) performance on two widely-used datasets. Specifically, it achieves 86.05% AP scores and 88.59% AUC scores on the XD-Violence and UCF-Crime datasets, outperforming all compared methods in the experiments.

2. Related work

2.1. Weakly supervised video anomaly detection and state space models

(Wu et al., 2024a) systematically synthesizes five categories of tasks and core methods in the field of deep learning-driven VAD. In WSVAD, most existing works (Ni et al., 2022; Zhou et al., 2023a) employ a multiple instance learning (MIL) framework, treating a video as a “bag” and its multiple segments as “instances” to infer frame-level anomalies from video-level annotations. Sultani et al. (2018) first propose a MIL ranking framework for WSVAD. Subsequent studies attempt to capture temporal dependencies utilizing self-attention mechanisms and Transformer models. For example, Tian et al. (2021) propose a VAD method that uses self-attention mechanisms and dilated convolutions to learn feature magnitudes and capture temporal dependencies. Zhong et al. (2019) propose a method based on a graph convolution network (GCN) to capture the similarity and temporal consistency across frames. Zhou et al. (2023a) propose a Transformer-based local-global multi-head self-attention module that generates more expressive embeddings to capture temporal dependencies in videos. Zhao et al. (2025) propose a VAD method based on a multi-scale temporal proxy mechanism. This approach employs a multi-scale temporal attention module and a temporal proxy mechanism to effectively capture temporal contextual information from video segments. These Transformer-based methods suffer from quadratic complexity growth with video sequence length.

State space models (SSMs) have recently emerged as a primary alternative to Transformer, garnering significant attention from researchers. Gu and Dao (2023) first propose the Mamba module, which not only establishes long-range dependencies but also dynamically and selectively updates the state based on the input. Such a design enables the module to achieve modeling capabilities comparable to those of Transformers while maintaining computational efficiency. Zhu et al. (2024) propose a general-purpose visual model, Vision Mamba, which employs a bidirectional state space model to learn visual representations efficiently. Ahamed et al. (Ahamed & Cheng, 2024) present Timemachine, a pure SSM-based model composed of four Mamba blocks for long-term time-series forecasting. In the VAD field, Lyu et al. (2025) propose a VAD method based on Mamba, VADMamba, which combines optical flow reconstruction and frame prediction to achieve efficient WSVAD.

In our work, LGM-Mamba innovatively integrates Mamba with Transformer and other components, leveraging their complementary strengths to achieve comprehensive modeling of temporal sequences.

2.2. Vision-language pre-training models and video caption

Vision-language pre-training models have recently attracted significant attention in various fields. As one of the most iconic achievements in this field, CLIP has exceptional performance in tasks such as dense prediction (Zhou et al., 2023b) and image classification (Zhou et al., 2022). Some work has successfully extended CLIP to the VAD field. For example, Lv et al. (2023) propose an Unbiased MIL (UMIL) framework using the visual features from CLIP. UMIL reduces context bias by clustering ambiguous segments and learning invariant features to improve accuracy. Similarly, Joo et al. (2023) propose a Temporal Self-Attention method that leverages CLIP’s visual features to extract discriminative representations for anomaly detection. While these methods can only detect whether the video is abnormal, Wu et al. (2024b) propose Vad-CLIP. The model combines components like an adapter, a prompt mechanism, and MIL-Align to capture temporal dependencies and optimize visual-textual alignment. However, these methods ignore the high-level semantic information hidden in the video.

Video caption technology can automatically parse the content and generate a natural language description that matches the scene, action, or event in the video. Recently, inspired by Transformer models (Lee & Toutanova, 2018) in the field of natural language processing, the computer vision community has been actively exploring the application

of these models in various downstream tasks. For example, Lin et al. (2022) propose an end-to-end video captioning approach that directly processes raw video frames using a video Transformer. In the VAD field, TEVAD (Chen et al., 2023) effectively captures the semantic information of abnormal events by extracting caption features, thus improving the interpretability of the detection results. LAVAD (Zanella et al., 2024) leverages pre-trained VLMs and LLMs for training-free video anomaly detection: it cleans frame captions via cross-modal similarity, generates temporal summaries with LLMs to estimate initial anomaly scores, and refines scores by aggregating semantically similar frames’ scores.

In our work, TrCLIP-VAD innovatively incorporates caption features into CLIP-based training. The framework addresses the critical issue of image-text asymmetry through an innovative text rewriting strategy, while simultaneously mitigating the risk of caption data overfitting.

3. Method

3.1. Overall framework

Fig. 2 shows an overview of TrCLIP-VAD framework. Given the training video $\mathcal{V}$, the video enters two independent branches to extract visual and caption features. In the text branch, we employ the SwinBERT model to generate dense captions for videos and further utilize the LLaMA-7b model to rewrite these captions. We randomly select a caption and transform it into sentence embeddings, and then capture the temporal relationship between captions through multi-scale temporal networks to obtain rich text features F_t . In the visual branch, we use the CLIP’s image encoder to extract visual features F_v . The LGM-Mamba module processes F_v to enhance its expressive ability. The fused feature F_{vt} is obtained by concatenating F_v and F_t . Then F_{vt} is sent to a coarse-grained binary classification branch (C-branch) and a fine-grained multiple classification branch (F-branch). By applying the Top-k mechanism, the C-branch identifies the video frames that are most likely to contain abnormal behaviors. In the F-branch, text labels such as abuse, riot, and fighting are first input into the CLIP text encoder and converted into corresponding class embeddings. Then by calculating the similarity between these class embeddings and the fused features F_{vt} , an alignment map $M \in \mathbb{R}^{n \times m}$ is generated, where n represents the number of video frames and m represents the number of abnormal types. This enables fine-grained video anomaly detection.

3.2. Generating caption features

3.2.1. Generating source captions

Although some researches (Krishna et al., 2017; Wang et al., 2021) are devoted to generating dense captions for video, the performance of these models is usually less satisfactory than that of single caption generation models, especially for the dense caption model. It is a challenge to identify the number of important events in the video. After comparison, we find that the SwinBERT (Lin et al., 2022) single caption model achieves good performance. Because of this, we chose SwinBERT to generate captions. A single video snippet typically has fewer frames, making it challenging to generate meaningful captions. SwinBERT employs a sliding window strategy, calculating the caption of 64 consecutive frames every 16 frames. This approach effectively reduces information omission and ensures that important events in the video can be captured. In addition to performance, another reason for selecting SwinBERT to generate captions is that it uses Video Swin Transformer to extract visual features, rather than the CLIP image encoder used in visual branch of TrCLIP-VAD. Diverse network architectures facilitate the learning of distinct representations, thereby enhancing anomaly detection performance.

3.2.2. Rewrite caption with LLaMA

Considering the high financial and time cost of rewriting a lot of text samples using closed-source models such as Bard or ChatGPT, we

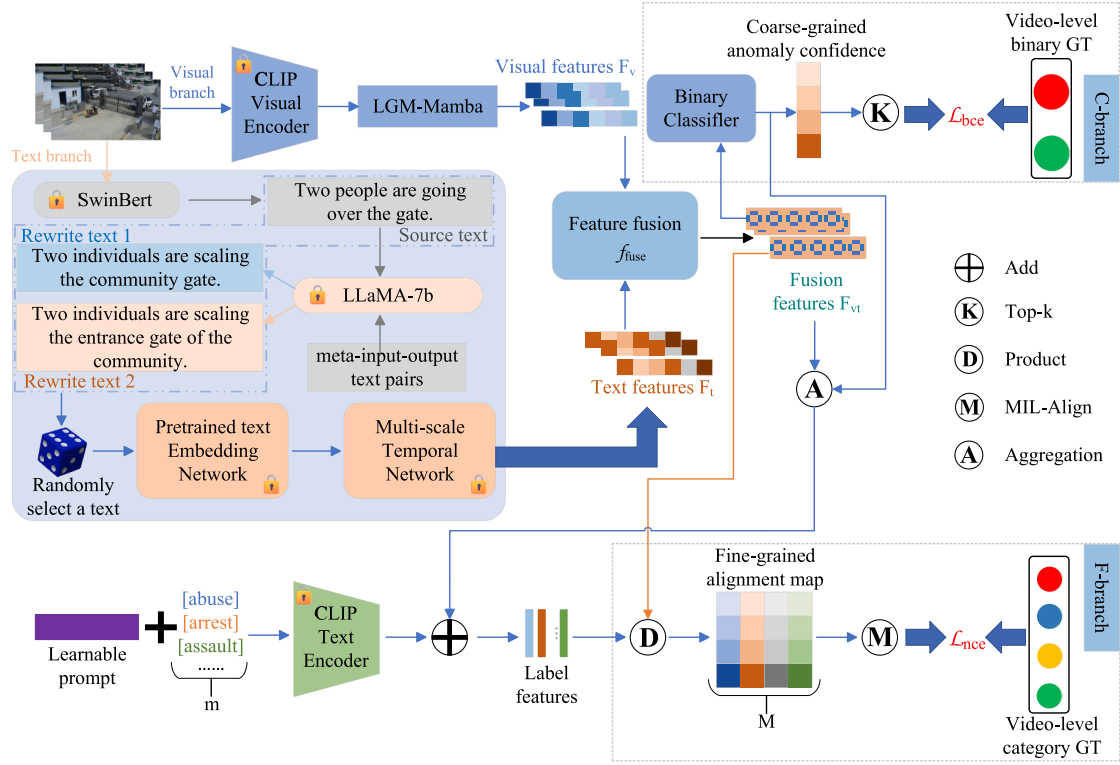


Fig. 2. The overview of the proposed TrCLIP-VAD. It consists of several components, i.e., visual branch, text branch, C-branch (coarse-grained branch), and F-branch (fine-grained branch).



Fig. 3. Illustration of generating meta-input-output pairs using ChatGPT: We first sample some examples from source captions. Subsequently, we employ instructions such as “Rewrite this image caption” to guide ChatGPT in generating target captions.

decide to adopt LLaMA. It is an open-source large-scale language model and is famous for its stable performance in text tasks. Even without fine-tuning specific instructions, LLaMA still shows excellent ICL ability, which makes it highly suitable for text rewriting tasks on any image-text dataset. Following the previous work (Fan et al., 2023), before proceeding with the rewriting, we need to generate several rewriting examples (meta-input-output text pairs) to guide LLaMA in rewriting the entire caption dataset. We adopt two strategies to generate these text pairs: rewriting with ChatGPT and by a human. **Rewriting with ChatGPT:** We randomly sample captions from the generated source caption datasets and feed them into ChatGPT. We prompt ChatGPT to generate the target text by using the prompt “Rewrite the image caption vividly while keeping it under 30 words.” The process is shown in Fig. 3. **Rewritten by human:** After randomly sampling the captions, we invite professional annotators to rewrite them based on the content of the corresponding images. Subsequently, we use LLaMA’s ICL ability to rewrite all text items on the image-text datasets in two different ways by using two meta-input-output text pairs as references.

To ensure logical event continuity in captions, we propose a temporal coherence preservation mechanism. As formalized in Algorithm 1, each sentence is anchored to its source timestamp t_i . Following independent sentence-level rewriting, we reconstruct the full caption strictly adhering to the original temporal sequence. For each rewritten sentence r_i , the system performs the following:

1. **Temporal Connector Audit:** Counting missing temporal logic words (e.g., ‘then’, ‘suddenly’) against the original sentence s_i , triggering rejection if Missing Connectors > 2 .
2. **Action Sequence Verification:** Ensuring original verb sequence $V_s = [v_1, v_2, \dots, v_n]$ is preserved in r_i with order invariance;
3. **Cross-Sentence Causality Scoring:** Assess the logical flow between the current sentence and the preceding one, and evaluate whether r_i logically follows from r_{i-1} . If the score is below 0.7, verification fails.

If any tier fails, the conservative rewriting fallback mechanism will be activated. This mechanism restricts modifications to only synonym substitution, while freezing the sentence structure and core entities. Such a design ensures causal integrity.

Algorithm 1 Text rewriting algorithm.

```

1: Input: Source captions  $C = [S_{t_1}, S_{t_2}, \dots, S_{t_n}]$ 
2: Output: Rewritten paragraph  $R = [R_{t_1}, R_{t_2}, \dots, R_{t_n}]$ 
3: for each sentence  $S_{t_i}$  in  $C$  do
4:   Independently rewrite  $S_{t_i} \rightarrow R_{t_i}$  using LLaMA
5: end for
6: Reorganize in the original order  $R = [R_{t_1}, R_{t_2}, \dots, R_{t_n}]$ 
7: if Missing Connectors  $> 2$  or Verb sequence  $V_s$  is preserved or
   Causality score  $\leq 0.7$  then
8:   conservative rewriting fallback
9: else
10:  return  $R$ 
11: end if

```

After obtaining three kinds of captions, we randomly select one of them and use SimCSE (Gao et al., 2021) to generate its sentence embedding. Then we connect these embedded features with the visual features to construct the fusion features.

3.3. LGM-Mamba

Inspired by Ahamed and Cheng (2024), Chen et al. (2023), Wu et al. (2024b), we design the LGM-Mamba module, which is one of the core parts of the TrCLIP-VAD framework. As shown in Fig. 4, LGM-Mamba is an innovative spatial-temporal adapter module specifically designed to process video time series.

3.3.1. Local and global temporal adapter (LGT adapter)

Local Temporal Adapter (LTA) employs a Transformer encoder layer to capture short-term dynamic changes in video. Unlike the traditional Transformer encoder layer, LTA limits the attention range of self-attention to a local window. This method can not only make the model focus more on local information and short-term dependencies in the video, but also reduce the computational complexity.

Global Temporal Adapter (GTA) uses a lightweight graph convolution network (GCN) (Zhong et al., 2019) module to capture global temporal dependencies. GCN captures global temporal dependencies by considering feature relative distance similarity, which can be presented as follows:

$$V_g = \text{gelu}([\text{Softmax}(A_s); \text{Softmax}(A_d)])V_l Z, \quad (1)$$

where A_s and A_d are the adjacency matrices of feature similarity and position distance, respectively. V_l is the visual feature processed by the local module. Z is the only learnable weight matrix used for transforming the feature space. The feature similarity matrix A_s is computed using the cosine similarity between the visual feature vectors, which is presented as follows:

$$A_s = \frac{V_l V_l^T}{\|V_l\|_2 \cdot \|V_l\|_2}. \quad (2)$$

The positional distance matrix A_d captures long-range dependencies within video sequences by computing the positional distances between each pair of frames, it can be presented as follows:

$$A_d(i, j) = \frac{-|i - j|}{\sigma}, \quad (3)$$

where i and j represent the i^{th} and j^{th} frames, respectively. The adjacency between these frames is entirely determined by their relative temporal positions. The hyperparameter σ determines the range of influence of the distance relation.

3.3.2. Mamba

Here, we integrate four Mamba blocks, the structure of each Mamba is illustrated in Fig. 5. It captures the long- and short-term dependencies in multi-variable time series using the state space model, while maintaining linear scalability and low memory usage.

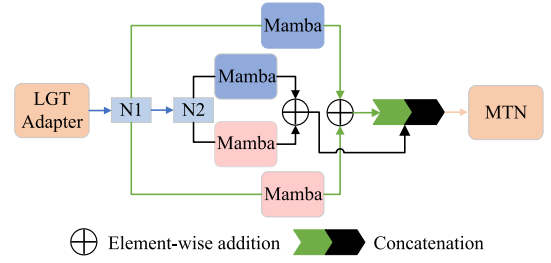


Fig. 4. The framework of our proposed LGM-Mamba module.

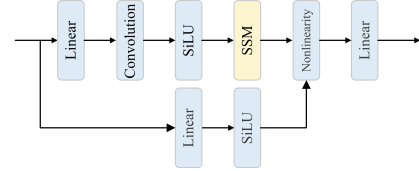


Fig. 5. The framework of the Mamba module.

Before processing the input sequence with Mamba, V_g is first transformed by two embedding layers N_1 and N_2 through the following operations:

$$n^{(1)} = N_1(V_g), \quad n^{(2)} = N_2(D(n^{(1)})), \quad (4)$$

where D denotes the dropout operation. The embedding layer N_1 transforms the shape of V_g from BSF to BSn_1 , while N_2 further projects it to BSn_2 , where B , S and F correspond to batch size, sequence length and feature dimension. n_1 and n_2 represent the output feature dimensions of the two embedding layers, and $n_1 > n_2$. The converted data $n^{(1)}$ and $n^{(2)}$ are processed by two Mamba modules to learn more comprehensive representations. The Mamba pair that handles $n^{(1)}$ is called the external Mamba pair, and the Mamba pair that handles $n^{(2)}$ is called the internal Mamba pair. The features processed by both internal and external Mamba pairs are concatenated and then fed into the MTN for further processing.

3.3.3. Multi-scale temporal network

Inspired by TEVAD (Chen et al., 2023), we use a multi-scale temporal network (MTN) (Tian et al., 2021) based on pyramid dilated convolution (PDC) (Liu et al., 2018) and non-local block (NLB) (Wang et al., 2018) to process visual and text features, respectively. Both visual MTN and text MTN use three layers of PDC blocks and one layer of NLB. Specifically, the input visual features are extracted from local to global spatio-temporal features through PDC, supplemented by NLB to enhance long-range dependency modeling. At the same time, the text features are processed by the temporal encoder of the same structure to maintain structural consistency between modalities.

3.4. Learnable prompt

In WSVAD, prompts are often concise words or short phrases that fail to summarize anomalous events adequately. To enhance the robust transferability of text embedding, we follow the setup in Wu et al. (2024b) and incorporate learnable prompts into the original class embedding. Specifically, we initially convert the raw textual labels into class tokens using the CLIP tokenizer:

$$T_{\text{init}} = \text{Tokenizer}(L), \quad (5)$$

where L represents a prompt, such as fighting, shooting, traffic accidents, etc. The class token T_{init} is placed in the middle of the sequence $\{c_1, \dots, c_n\}$. The input to the text encoder is represented as follows:

$$T_p = \{c_1, \dots, T_{\text{init}}, \dots, c_n\}, \quad (6)$$

where T_p is the input of the CLIP text encoder.

3.5. Objective function

In the C-branch, we follow the Top-K mechanism (Tian et al., 2021) that selects the K video segments with the highest anomaly confidence scores to serve as video-level prediction. We compute the classification loss $\mathcal{L}_{bce}$ between the video-level predictions and the ground-truth labels using binary cross-entropy.

In the F-branch, we adopt the MIL-Align mechanism proposed by Wu et al. (2024b), which captures the similarity between frame-level visual features and various abnormal categories. For each row, MIL-Align selects the top k features that are most similar to the corresponding category and calculates their average similarity. It is used to measure the consistency between a video and a specific category. MIL-Align aims to minimize the similarity between the visual feature and its irrelevant labels, while maximizing similarity with correct labels. We first calculate the multi-category prediction, the specific steps are as follows:

$$p_i = \frac{\exp\left(\frac{q_i}{\tau}\right)}{\sum_j \exp\left(\frac{q_j}{\tau}\right)}, \quad (7)$$

where τ denotes the temperature hyperparameter for scaling, q_i and p_i represent the raw and final score that the model predicts for the i^{th} class. Finally, the alignment loss $\mathcal{L}_{nce}$ can be calculated by the cross-entropy.

Besides $\mathcal{L}_{bce}$ and $\mathcal{L}_{nce}$, there is also a contrast loss $\mathcal{L}_{cts}$. It calculates the contrast loss $\mathcal{L}_{cts}$ by computing the cosine similarity between the embeddings of the normal class and those of other classes. The calculation method is as follows:

$$\mathcal{L}_{cts} = \sum_j \max\left(0, \frac{x_n^\top x_{aj}}{\|x_n\|_2 \cdot \|x_{aj}\|_2}\right), \quad (8)$$

where x_n and x_a are the normal and abnormal class embeddings, respectively.

Overall, the final total objective of TrCLIP-VAD is as follows:

$$\mathcal{L} = \mathcal{L}_{bce} + \mathcal{L}_{nce} + \lambda \mathcal{L}_{cts}. \quad (9)$$

Here, λ is a hyperparameter that is used to balance the weights of losses.

4. Experiments

4.1. Datasets and evaluation metrics

Datasets. We conduct experiments on two well-known WSVAD datasets: UCF-Crime (Sultani et al., 2018) and XD-Violence (Wu et al., 2020). The UCF-Crime dataset consists of 1900 untrimmed surveillance video clips (128 h total), covering 13 categories of real-world anomalous events, such as fighting and arson. The XD-Violence dataset focuses on violent behavior, containing 4754 video clips (217 h total). It contains six categories of highly dynamic violent events, such as explosions and car accidents.

Evaluation Metrics. For coarse-grained WSVAD, we follow the previous work (Feng et al., 2021; Wu et al., 2020) using the frame-level AP score for XD-Violence dataset and the frame-level AUC score for UCF-Crime dataset. For fine-grained WSVAD, we follow the settings in Wu et al. (2022), which uses the mean average precision (mAP) values under different intersection over union (IoU) thresholds. Specifically, we select the IoU threshold range from 0.1 to 0.5, and calculate the mAP in increments of 0.1. Additionally, we calculate the average mAP (AVG) from 0.1 to 0.5. To assess the semantic consistency between rewritten captions and their source captions, we employ BLEU (Papineni et al., 2002), ROUGE (Lin, 2004), and METEOR (Banerjee & Lavie, 2005) metrics, as well as manual evaluation.

4.2. Implementation details

For visual and label features, we use image and text encoders from the pre-trained CLIP model (ViT-B/16). The dimension of the visual features is $d_v = 512$, and the dimension of the label features is $d_l = 1280$.

For the caption features, we use SwinBERT that has been pre-trained on the VATEX (Wang et al., 2019) dataset to generate image captions, and employ the LLaMA-7B model to generate two distinct rewrites for each caption. Using a server equipped with eight A40 GPUs, generating one round of rewrites takes approximately 13 h for the XD-Violence dataset and 11 h for the UCF-Crime dataset. The dimension of caption features is $d_c = 768$. We set the hyperparameter τ in Eq. (7) as 0.07. N1 and N2 in LGM-Mamba are set to 512 and 256, respectively. In Eq. (9), the parameter λ is assigned a value of 1×10^{-4} on XD-Violence and 1×10^{-1} on UCF-Crime dataset. TrCLIP-VAD is trained on a single NVIDIA RTX A40 GPU using PyTorch. We use AdamW as the optimizer with a batch size of 64. On XD-Violence and UCF-Crime datasets, the total number of epochs is set to 10, and the learning rate is set to 6×10^{-6} and 7×10^{-5} .

4.3. Comparison with state-of-the-art methods

Coarse-grained WSVAD Results. We compare our proposed TrCLIP-VAD method with SOTA methods on XD-Violence and UCF-Crime datasets. The top two results in each category are marked in **bold** and underline. The results are summarized in Tables 1 and 2. Based on the level of supervision used during training, we categorize the methods into two groups: semi-supervised and weakly-supervised.

As shown in Tables 1 and 2, TrCLIP-VAD achieves 86.05% and 88.59% on XD-Violence and UCF-Crime, surpassing all other methods. Notably, LAVAD represents a novel training-free paradigm that leverages pre-trained VLMs and LLMs to avoid task-specific training, but its performance is still outperformed by TrCLIP-VAD. This indicates that our weakly-supervised framework can better capture task-specific discriminative features compared to training-free semantic reasoning. Beyond outperforming training-free methods like LAVAD, TrCLIP-VAD extracts more comprehensive features than VADMamba – a pure Mamba-based model that often struggles to capture local information effectively. In addition, AVVD relies entirely on fine-grained class labels, yet its performance significantly lags behind TrCLIP-VAD. This indicates that using only fine-grained labels does not necessarily enhance model performance. In contrast, the caption features of TrCLIP-VAD compensate for the lack of high-level semantic information in visual features.

Fine-grained WSVAD Results. We compare TrCLIP-VAD with prior work VadCLIP (Wu et al., 2024b), Sultani et al. (2018) and AVVD (Wu et al., 2022) in Tables 3 and 4. AVVD first proposes fine-grained WSVAD, and VadCLIP re-implements it using CLIP's visual features. In the fine-grained WSVAD task, whether on the XD-Violence or UCF-Crime dataset, the mAP of TrCLIP-VAD is significantly higher than other methods under all IoU thresholds. Especially when $\text{IoU} = 0.1$, TrCLIP-VAD achieves 42.64% and 16.01%, respectively, representing increases of 5.61% and 4.29%. The average mAP values are 30.63% and 9.17%, representing increases of 5.79% and 2.49%. These results are the best among all the comparison methods. The text-visual co-enhancement and multi-scale temporal modeling of TrCLIP-VAD establish new SOTA in fine-grained WSVAD.

4.4. Ablation studies

We conduct extensive ablation studies on two datasets to assess the contributions of the core components of the TrCLIP-VAD method. Additionally, we conduct an in-depth assessment of the quality of rewritten captions.

4.4.1. Effectiveness of LGM-Mamba

The results of the ablation study in Table 5 demonstrate that every component of the LGM-Mamba module, including LTA, GTA, Mamba, and MTN, contributes positively to VAD performance. Notably, when these components operate in concert, the model achieves its best performance on the XD-Violence and UCF-Crime datasets. This indicates that by integrating local perception, global modeling, and multi-scale analysis capabilities, the LGM-Mamba module can effectively capture

Table 1
Coarse-grained comparisons on XD-violence dataset.

Category	Methods	Source	AP(%)
Semi	SVM baseline	NIPS	50.00
	OCSVM (Schölkopf et al., 1999)	NIPS	28.63
	Hasan et al. (2016)	CVPR	31.25
Weak	Ju et al. (2022)	ECCV	76.57
	Sultani et al. (2018)	CVPR	75.18
	Wu et al. (2020)	ECCV	80.00
	RTFM (Tian et al., 2021)	ICCV	78.27
	AVVD (Wu et al., 2022)	TMM	78.10
	DMU (Zhou et al., 2023a)	AAAI	82.41
	CLIP-TSA (Joo et al., 2023)	ICIP	82.17
	TPWNG (Yang et al., 2024)	CVPR	83.68
	VadCLIP (Wu et al., 2024b)	AAAI	84.51
	BN-WVAD (Zhou et al., 2024)	TCSVT	85.26
	VADMamba (Lyu et al., 2025)	ICME	83.62
	LAVAD (Zanella et al., 2024)	CVPR	85.36
	TrCLIP-VAD(Ours)	—	86.05

Table 2
Coarse-grained comparisons on UCF-Crime dataset.

Category	Methods	Source	AUC(%)
Semi	SVM baseline	NIPS	50.00
	OCSVM (Schölkopf et al., 1999)	NIPS	63.20
	Hasan et al. (2016)	CVPR	39.43
Weak	Ju et al. (2022)	ECCV	84.72
	Sultani et al. (2018)	CVPR	84.14
	Wu et al. (2020)	ECCV	84.57
	RTFM (Tian et al., 2021)	ICCV	85.66
	AVVD (Wu et al., 2022)	TMM	82.45
	DMU (Zhou et al., 2023a)	AAAI	86.75
	UMIL (Lv et al., 2023)	CVPR	86.75
	CLIP-TSA (Joo et al., 2023)	ICIP	87.58
	LAVAD (Zanella et al., 2024)	CVPR	80.28
	TPWNG (Yang et al., 2024)	CVPR	87.79
	BN-WVAD (Zhou et al., 2024)	TCSVT	87.24
	VadCLIP (Wu et al., 2024b)	AAAI	88.02
	VADMamba (Lyu et al., 2025)	ICME	87.23
	TrCLIP-VAD(Ours)	—	88.59

Table 3
Fine-grained comparisons on XD-Violence dataset.

Methods	mAP@IOU(%)					
	0.1	0.2	0.3	0.4	0.5	AVG
Random	1.82	0.92	0.48	0.23	0.09	0.71
Sultani et al. (2018)	22.72	15.57	9.98	6.20	3.78	11.65
AVVD (Wu et al., 2022)	30.51	25.75	20.18	14.83	9.79	20.21
VadCLIP (Wu et al., 2024b)	37.03	30.84	23.38	17.90	14.31	24.70
TrCLIP-VAD(Ours)	42.37	36.09	30.21	24.61	19.86	30.63

Table 4
Fine-grained comparisons on UCF-Crime dataset.

Methods	mAP@IOU(%)					
	0.1	0.2	0.3	0.4	0.5	AVG
Random	0.21	0.14	0.04	0.02	0.01	0.08
Sultani et al. (2018)	5.73	4.41	2.69	1.93	1.44	3.24
AVVD (Wu et al., 2022)	10.27	7.01	6.25	3.42	3.29	6.05
VadCLIP (Wu et al., 2024b)	11.72	7.83	6.40	4.53	2.93	6.68
TrCLIP-VAD(Ours)	16.01	11.74	9.40	5.68	3.01	9.17

and leverage complex temporal features within video data, thereby enabling more accurate anomaly detection.

4.4.2. Effectiveness of TrCLIP-VAD

Table 6 shows the impact of different input combinations on model performance for the XD-Violence and UCF-Crime datasets. When visual features are combined with each of the other three components, there

Table 5
Effectiveness of the LGM-Mamba module on XD-Violence and UCF-Crime.

LTA	GTA	Mamba	MTN	XD-Violence		UCF-Crime	
				AP(%)	AVG(%)	AUC(%)	AVG(%)
✓	✓	✓	✓	72.22	15.64	77.46	3.12
				81.18	18.41	82.17	6.10
				82.54	16.76	82.25	6.11
✓	✓	✓	✓	81.78	22.55	86.78	5.56
				80.96	23.24	87.22	5.23
				84.51	24.07	88.02	6.68
✓	✓	✓	✓	83.22	25.66	87.12	7.43
				84.54	25.75	88.10	8.02
				84.60	25.88	88.23	8.34
✓	✓	✓	✓	85.29	28.15	88.59	8.94

Table 6
Effectiveness of TrCLIP-VAD on XD-Violence and UCF-Crime. Here, S-caption is the source caption and R-caption is the rewritten caption.

LGM-Mamba	S-caption	R-caption	XD-Violence		UCF-Crime	
			AP(%)	AVG(%)	AUC(%)	AVG(%)
✓	✓	✓	72.22	15.64	77.46	3.12
			83.54	26.07	87.25	6.11
			84.22	26.55	87.52	8.28
✓	✓	✓	85.29	28.15	88.23	8.94
			85.74	28.75	88.25	9.01
			86.05	30.74	88.59	9.39

is a varying degree of enhancement. Notably, combining LGM-Mamba with S-caption yields further performance improvement. Ultimately, the combination of LGM-Mamba and R-caption proves to be the most effective. This shows that the processing of LGM-Mamba and rewriting of the caption significantly enhance model performance on two datasets, especially in the fine-grained WSVAD task. In conducting a fine-grained analysis, the model must identify and distinguish subtle differences. The caption rewriting helps the model understand these subtle differences better, thereby enabling it to achieve more accurate classification and prediction in fine-grained recognition tasks.

4.4.3. Semantic temporal consistency between R-captions and S-captions

To evaluate whether the R-captions retain the semantics of the S-captions, we randomly select 500 S-captions and their corresponding rewritten captions from XD-Violence and UCF-Crime datasets. We conduct manual evaluation and use automate metrics (BLEU Papineni et al., 2002 and ROUGE Lin, 2004). The results are shown in Table 7.

Manual evaluation: We invite ten professional annotators to evaluate each pair of texts independently. The evaluation criteria are as follows:

A. Complete consistency: The R-captions completely retain the semantics of the S-captions.

B. Inconsistency: There are significant semantic differences between the R-captions and the S-captions.

C. Partial consistency: The R-captions retain most of the semantics of the S-captions, but little information is lost or changed.

The annotator evaluate each pair of texts and record the evaluation results. Finally, we calculate the proportion of each evaluation result. The evaluation results show that 85% of the rewritten texts are rated as completely consistent, 13% as partially consistent, and only 2% as inconsistent. This shows that our rewriting strategy effectively preserves the original semantics.

Automated metrics: To further confirm the semantic retention of rewritten captions, we calculate the scores of automation indicators BLEU, ROUGE, and METEOR.

These metrics assess the n-gram overlap and longest common subsequence similarity between the R-captions and S-captions, respectively. The results from these automated evaluations corroborate the manual

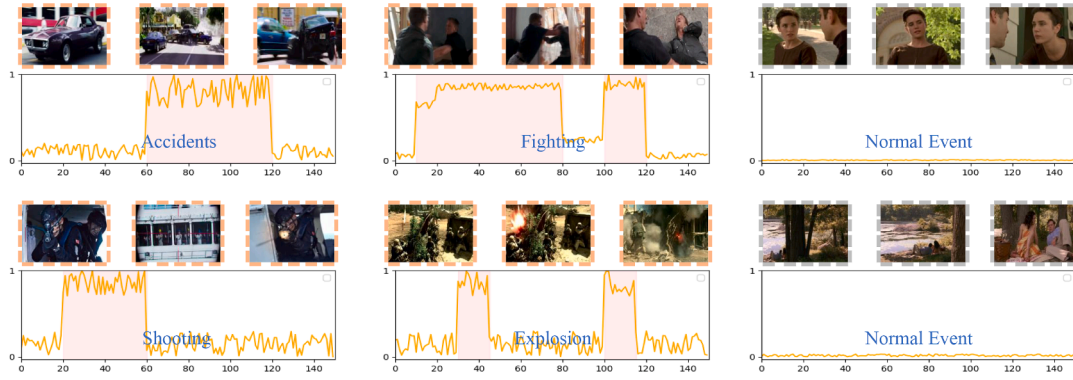


Fig. 6. Qualitative results of coarse-grained WSVAD. The orange curve represents the anomaly score and the pink area indicates the temporal location of anomalies.

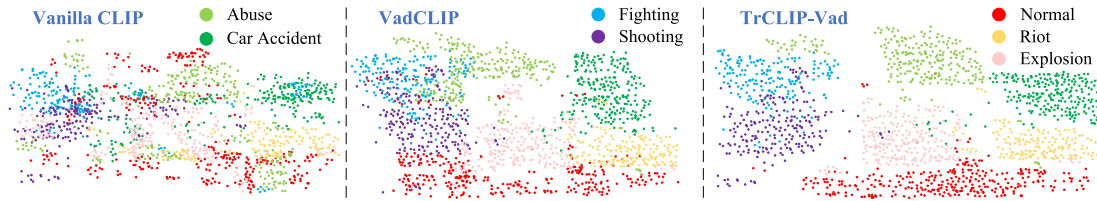


Fig. 7. t-SNE visualizations for XD-Violence dataset. Left: Raw CLIP features; Middle: VadCLIP features; Right: TrCLIP-VAD features.

Table 7

Quantitative evaluation of semantic consistency between R-captions and S-caption.

Methods	XD-Violence			UCF-Crime		
	BLEU	ROUGE	METEOR	BLEU	ROUGE	METEOR
Qwen	0.727	0.275	0.104	0.798	0.280	0.101
Bard	0.736	0.219	0.098	0.751	0.211	0.108
ChatGPT	0.785	0.298	0.110	0.758	0.257	0.105
LLaMA	0.808	0.301	0.121	0.814	0.314	0.116

Table 8

Efficiency comparison between our TrCLIP-VAD and VadCLIP on the XD-Violence dataset. We report training time per epoch, GPU memory usage, and total parameters.

Methods	XD-Violence		
	GPU Memory (MB)	Epoch (min)	Params (M)
VadCLIP	4873	2.5	159.11
TrCLIP-VAD (Our)	9707	4.7	191.41

Table 9

Efficiency comparison between our TrCLIP-VAD and VadCLIP on the UCF-Crime dataset. We report training time per epoch, GPU memory usage, and total parameters.

Methods	UCF-Crime		
	GPU Memory (MB)	Epoch (min)	Params (M)
VadCLIP	7327	6.9	162.26
TrCLIP-VAD (Our)	13,984	8.0	194.56

assessment findings, further validating the efficacy of our rewriting approach.

4.4.4. The efficiency of TrCLIP-VAD framework

We further conduct a comprehensive efficiency analysis to evaluate the computational overhead introduced by the proposed TrCLIP-VAD framework, with the results presented in Tables 8 and 9. As shown in Table 8, on the XD-Violence dataset, our TrCLIP-VAD consumes

9707 MB of GPU memory, requires 4.7 min per training epoch, and has a total of 191.41 million parameters. Compared with the VadCLIP baseline, the introduction of LLaMA for text rewriting brings a certain increase in computational cost: the GPU memory usage is approximately doubled, the training time per epoch increases by 88%, and the total parameters grow by 20.3%. Similarly, on the UCF-Crime dataset (Table 9), TrCLIP-VAD achieves 13984 MB of GPU memory, 8.0 min per epoch, and 194.56 million parameters, which also shows a moderate overhead compared to VadCLIP. Notably, despite the additional computational cost, the performance improvement brought by our text-rewriting enhanced CLIP training is significant, and the efficiency of TrCLIP-VAD remains within an acceptable range for practical video anomaly detection applications.

4.5. Qualitative analysis

Coarse-grained Qualitative Visualization. We define an anomaly score exceeding 0.5 as an anomaly event. As shown in Fig. 6, TrCLIP-VAD can accurately identify anomaly regions in different categories on two datasets. The model's anomaly score is close to 0 for normal cases and near 1 for anomalous cases.

Feature Discrimination Visualization. Fig. 7 illustrates the distribution of data points in two-dimensional space after dimensionality reduction using t-SNE, with seven different categories represented by distinct colors. It can be observed that the features of raw Vanilla CLIP are severely mixed across categories, failing to form effective distinguishing boundaries. After optimization by VadCLIP (Wu et al., 2024b), the category separation is slightly improved, but different classes (e.g., Fighting and Shooting) still show obvious overlap. In contrast, after the joint optimization of TrCLIP-VAD, the features of each category are clustered more tightly, and the boundaries between different classes (especially Normal, Explosion, and Abuse) become significantly clearer, demonstrating that our method effectively enhances the discriminability of WSVAD features.

5. Conclusions

In this work, we propose TrCLIP-VAD, an innovative method for WSVAD, which introduces a text rewriting branch to address the asymmetry

between video captions and visual features. To reduce the risk of over-fitting, we randomly select one caption from multiple versions and fuse it with visual features for anomaly detection. Subsequently, we design the LGM-Mamba module to strengthen the model's time series representation capabilities. Through experiments on two WSVAD datasets, we show the superior performance of TrCLIP-VAD and validate its effectiveness through comprehensive ablation studies. In the future, we plan to explore further training-free language-based VAD, aiming to provide more efficient and intelligent solutions for related fields.

CRedit authorship contribution statement

Shengjie Shen: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization; **Ziteng Guo:** Visualization, Formal analysis, Data curation; **Yahui Li:** Writing – original draft, Visualization, Data curation; **Liejun Wang:** Supervision, Project administration, Funding acquisition; **Zhiqing Guo:** Writing – review & editing, Validation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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